

First Year B.S. Honors (Session: 2024-25)

Department of Applied Mathematics, University of Dhaka

Course Title: Math Lab I (Mathematica), Course No.: AMTH 150

Assignment 1 (part A)

Name:

Roll:

Group:

1. Express each set in roster method and find the cardinality of each set.
 - a) The set of Natural numbers less than 4.
 - b) The set of odd Natural numbers less than 13.
 - c) The set of numbers less than 1000 which are both prime and Fibonacci.
2. Determine whether each statement is true or false.
 - a) $3 \in \{1,3,5,7\}$
 - b) $6 \in \{2,4,6,8\}$
 - c) $10 \in \{1,2,3, \dots, 16\}$
 - d) $11 \notin \{1,2,3, \dots, 16\}$
 - e) $4 \notin \{x | x \in \mathbb{N} \text{ and is even}\}$
3. Draw three circles using **Disk** command. Now show the following regions separately using **RegionPlot**
 - a) A b) $A \cup B$ c) $(A \cap B \cap C)$ d) $A \setminus B$ e) $B \cup C$ f) $A \cup B \cup C$
4. Let $A = \{c\}$, $B = \{a,b\}$, $C = \{b,d\}$ and $U = \{a,b,c,d,e,f\}$. Find $A \cup (B' \cap C')$ and $(A \cup B') \cap (A \cup C')$.
5. Let $A = \{1,3, 4, 5,6\}$, $B = \{2, 4, 7, 8\}$ and $U = \{1, 2, 3, 4, 5,6, 7,8,9,11\}$. Show that, $(A \cup B)' = A' \cap B'$.
6. A university surveyed 300 students about membership in three clubs: **Drama (D)**, **Music (M)**, and **Sports (S)**.
 - 150 are in **Drama**,
 - 140 are in **Music**,
 - 120 are in **Sports**,
 - 60 are in **both Drama and Music**,
 - 50 are in **both Drama and Sports**,
 - 40 are in **both Music and Sports**,
 - 25 are in **all three clubs**.
 - a. How many students are in **exactly one club**?
 - b. How many students are in **exactly two clubs**?
 - c. How many students are in **at least one club**?
 - d. How many students are in **none of the clubs**?

7. The cost per mile for a rented vehicle is 50 cents for the first 100 miles, 30 cents for the next 200 miles, and 20 cents for all miles in excess of 300 miles. Write a program that determines the total cost and the average cost per mile for a given number of miles.
8. Using the rules of divisibility to check whether each number is divisible by 2 & 3. Also find the prime factorization of these numbers.
 a) 6944 b) 7245 c) 21408
9. A prime number is an *emirp* if it becomes a different prime number when its digits are reversed. Determine whether or not each prime number is an *emirp*.
 a) 41 b) 43 c) 107 d) 113
10. Find the product of *gcd* of 48 and 72 and the *lcm* of 48 and 72. Compare this result to the product of 48 and 72. Write a conjecture based on your observation.
11. Determine whether the following statements are equivalent using truth table:
 i) If the sky is clear and it is after sundown, then we can see Saturn.
 ii) If the sky is not clear or it is not after sundown, then we cannot see Saturn.
12. Determine which, if any, of the three statements are equivalent using truth table
 i) The package was sent by Fedex, or the package was not sent by the United Parcel Service but the package arrived on time.
 ii) The package arrived on time, if and only if the package was sent by the Fedex or the package was not sent by the United Parcel Service.
 iii) If the package was not sent by Fedex, then the package was not sent by the United Parcel Service but the package arrived on time.
13. Write the following grading criteria in table with proper headings and alignment.

≥ 80	75-79	70-74	65-69	60-64	55-59	50-54	45-49	40-44	< 40
4.0	3.75	3.5	3.25	3.0	2.75	2.5	2.25	2.0	0.0

A student gets the following numbers in eight courses:

A101	A102	A103	A104	A105	A106	A107	A150
76	82	68	78	87	67	58	80

where A102 and A104 are four-credit courses, and the rest are three-credit courses. Write a program that calculates the CGPA.